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Summary

AI & Intelligent Systems Engineer-in-training with hands-on experience in Deep Learning, Machine Learning, CNN, RNN, ANN, and IoT projects. Delivered high-quality AI solutions, debugging and optimizing models from scratch, and presenting results in IEEE competitions and workshops.

Education

Helwan National University — Bachelor in Intelligent Systems Engineering
Expected Graduation: 2027 | GPA: 3.4

Relevant Coursework: AI, Machine Learning, Deep Learning, CNN, RNN, Embedded Systems, IoT Development

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Experience

HCIA AI v4 Certification — NTI (Jul-Aug 2025)

- Completed 80 hours intensive AI training covering Deep Learning, CNN, RNN, ANN, and practical ML projects.
- Developed AI Style Transfer final project, transforming 50 images into polished 4K output using PyTorch & TensorFlow Hub.
- Participated in 4 competitions with average model accuracy of 90%.
- Scored 984/1000 in the final HCIA exam.

IoT & Soft Skills Training — NTI (Ongoing, Aug-Sep 2025)

- 120-hour program: 90 hours IoT development and 30 hours soft skills.
- Built Smart Car Parking System: Python & Arduino solution detecting correct parking, triggering green/red/blue LEDs with ultrasound & buzzer alerts, achieving flawless real-time operation.

IEEE Student Branch Activities

- Developed and debugged CNN from scratch and CNN transfer learning models, achieving 91% accuracy.
- Presented models in IEEE workshops to audiences of 50+ peers, demonstrating practical AI applications.
- Hands-on experience with ANN and RNN models in competitions and team challenges.

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, HTML, CSS, SQL, MATLAB, Arduino, R, Bash, TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Seaborn, Matplotlib

Embedded Systems & IoT: Atmega32, Arduino IDE, Sensors (Ultrasonic, IR, Laser), Actuators, Real-time Data Acquisition

AI & Machine Learning: Deep Learning, Machine Learning, Style Transfer, Image Classification, Recommendation Systems, CNN (from scratch & transfer learning), RNN, ANN, Data Analysis

Tools & Platforms: Git, GitHub, Kaggle, Jupyter Notebook, Excel

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Soft Skills

- Team Collaboration on AI & IoT projects
- Debugging and Model Optimization
- Presentation & Technical Communication
- Project Planning & Execution
- Organizing tasks to meet deadlines
- Identifying and resolving issues
- Research and Continuous Learning

Projects

AI Style Transfer (NTI Final Project)

- Transformed 50+ real photos into stylized 4K images using PyTorch & TensorFlow Hub.
- Polished results using OpenCV for enhancement.

Embedded Smart Trash Bin (University Project)

- Classified metal vs other waste using 2 Atmega32 microcontrollers on 2 PCBs, with LCD & ultrasonic feedback.
- Automated lid operation and real-time bin fill percentage display, handling average-sized trash items accurately.

Stair Master with Arduino & Laser Sensor (University Project)

- Miniature stair-climbing device with 3 speed modes using laser detection.
- Optimized response time by 30% compared to initial prototype.

Smart Car Parking System (Team Project)

- Python & Arduino system detecting parking status and triggering LEDs with buzzer alerts.
- Reduced parking errors to 0 during testing and optimized user guidance in real-time.

Certifications

- Machine Learning Fundamentals — DataCamp
- Python Basics — MaharaTech - ITIMooca
- IEEE Student Branch Certification (Upcoming)

Languages

- Arabic (Native)
- English (Proficient)
- Italian (Basic)